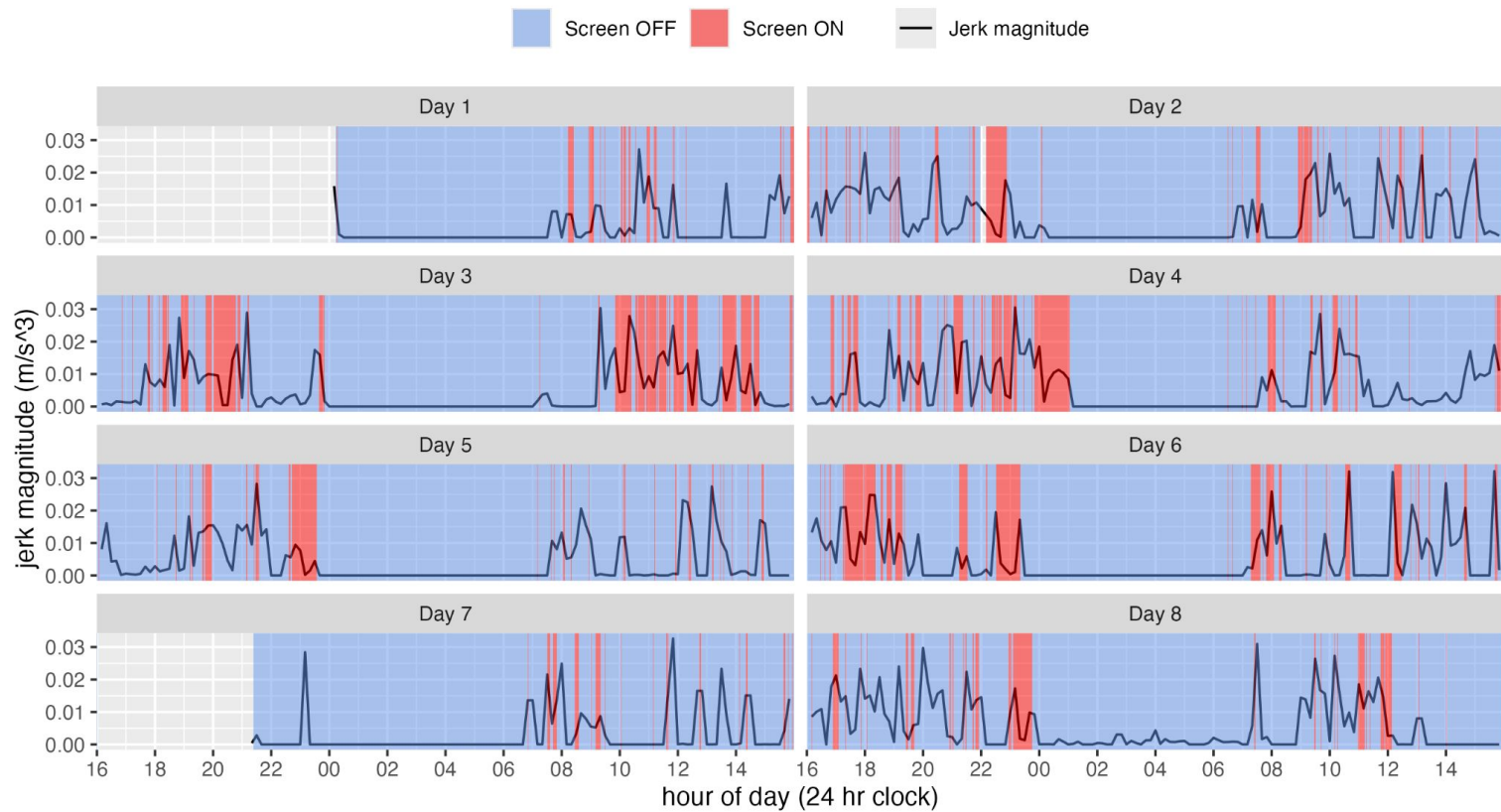


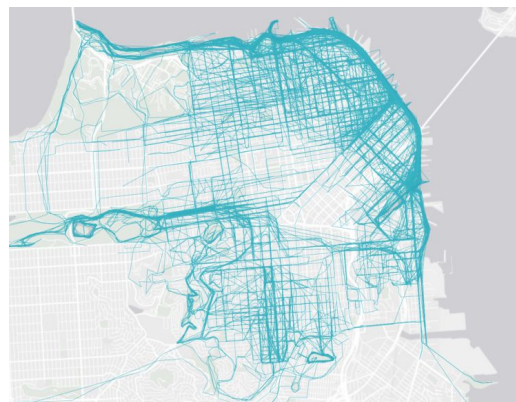
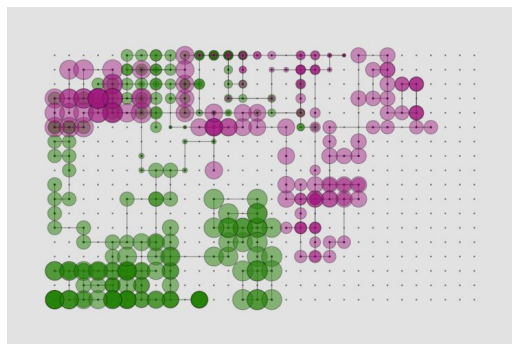
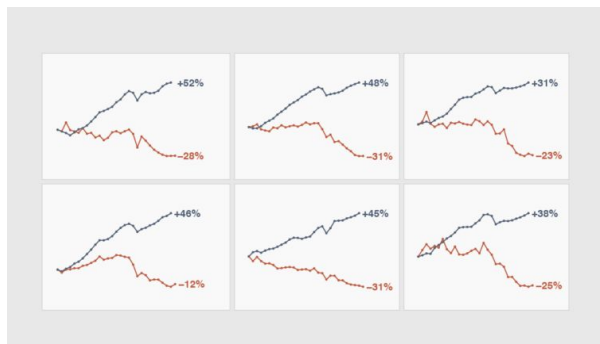
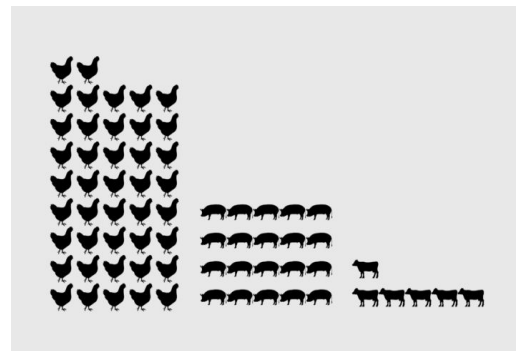
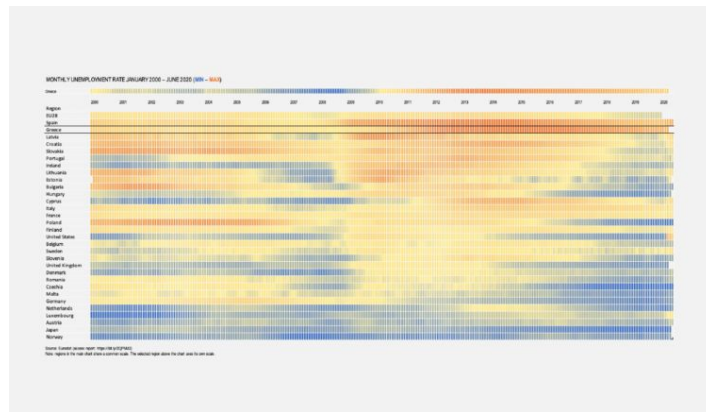
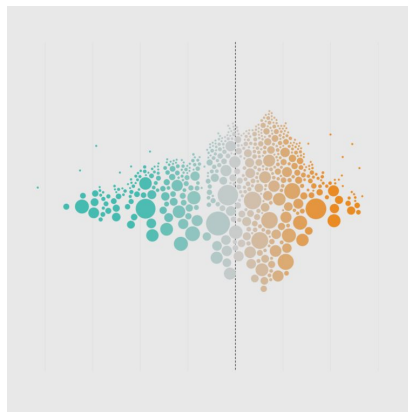
JULIANNA HSING | SEPTEMBER 22, 2026

data visualization & storytelling

let's take 10 seconds; what do you see?



Data viz ≠ pretty charts



What is data visualization?

The **graphical representation of data** to communicate **complex data relationships** and **data-driven insights** in a way that is **easy to understand**.

What is data visualization?

The **graphical representation of data** to communicate **complex data relationships** and **data-driven insights** in a way that is **easy to understand**.

“Well-designed presentations of interesting data are a matter of **substance**, of **statistics**, and of **design**”.

– Edward Tufte

Professor emeritus of political science, statistics, and computer science at Yale University

What is data visualization?

The **graphical representation of data** to communicate **complex data relationships** and **data-driven insights** in a way that is **easy to understand**.

“Well-designed presentations of interesting data are a matter of substance, of statistics, and of design”.

– Edward Tufte

Professor emeritus of political science, statistics, and computer science at Yale University



story and content
is clear, accurate,
and truthful

What is data visualization?

The **graphical representation of data** to communicate **complex data relationships** and **data-driven insights** in a way that is **easy to understand**.

“Well-designed presentations of interesting data are a matter of **substance**, of **statistics**, and of **design**”.

– Edward Tufte

Professor emeritus of political science, statistics, and computer science at Yale University



story and content
is clear, accurate,
and truthful



analysis is correct,
honest, and
properly scaled

What is data visualization?

The **graphical representation of data** to communicate **complex data relationships** and **data-driven insights** in a way that is **easy to understand**.

“Well-designed presentations of interesting data are a matter of **substance**, of **statistics**, and of **design**”.

- Edward Tufte

Professor emeritus of political science, statistics, and computer science at Yale University

story and content
is clear, accurate,
and truthful

analysis is correct,
honest, and
properly scaled

layout is clean and
easy for the viewer
to understand

Why do we visualize data?



Analytical visualization

- Goal: **exploration**
- We don't know what the data has to say yet
- Use charts to help **us** understand the data
- **We are both** the **maker** and the **consumer** of the charts



Presentative visualization

- Goal: **communication**
- We know what the data have to say
- Use charts to help **others** understand the data
- **We are the maker** and **our audience is the consumer** of the charts

Why do we visualize data?



Analytical visualization

- Goal: **exploration**
- We don't know what the data has to say yet
- Use charts to help **us** understand the data
- **We are both** the **maker** and the **consumer** of the charts



Presentative visualization

- Goal: **communication**
- We know what the data have to say
- Use charts to help **others** understand the data
- **We are the maker** and **our audience is the consumer** of the charts

3 key questions

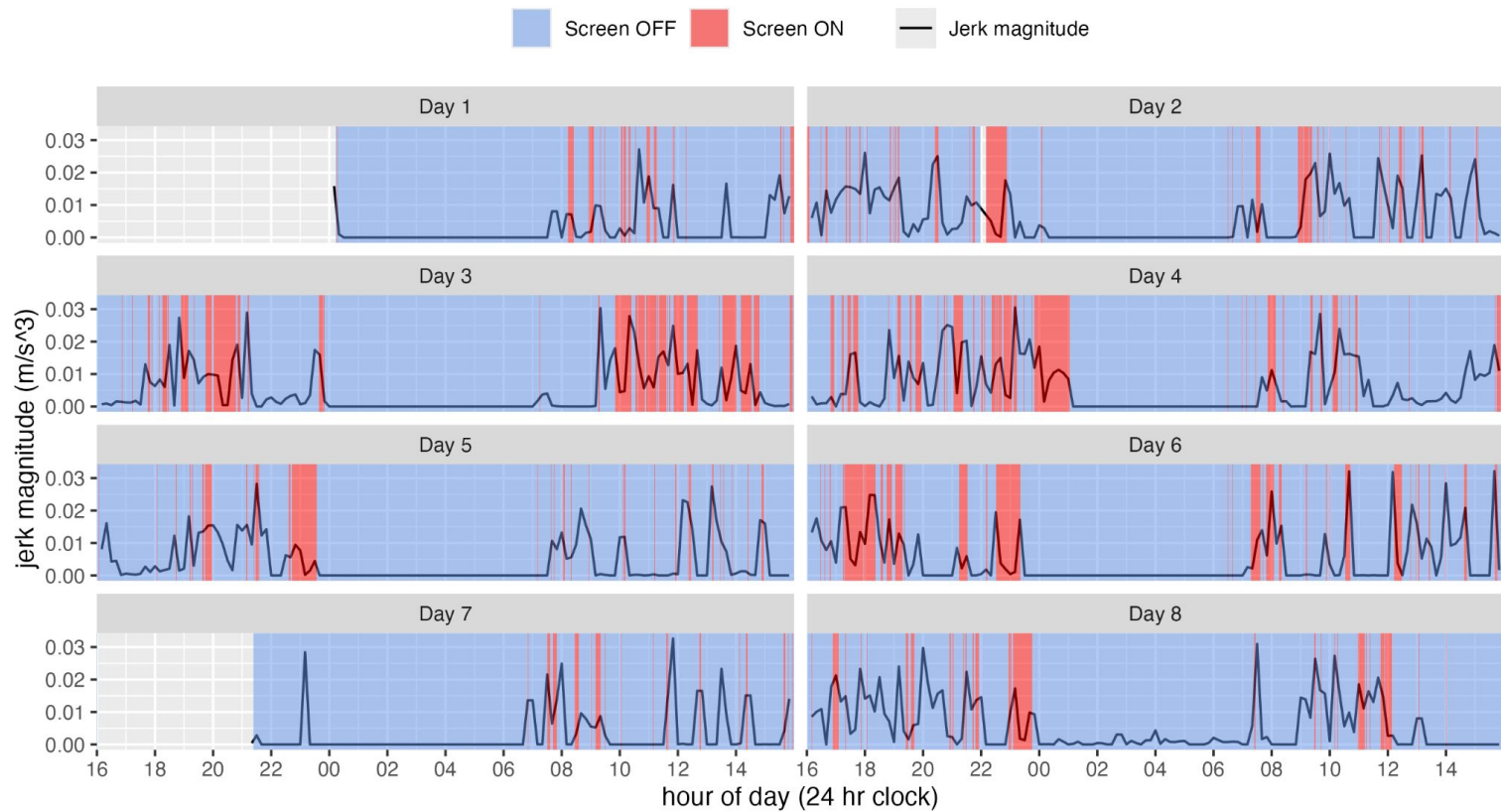
1. Who is my **audience**?

3 key questions

1. Who is my **audience**?
2. What **question** am I trying to answer?

3 key questions

1. Who is my **audience**?
2. What **question** am I trying to answer?
3. What **story** do I want people to walk away with?



3 key questions

1. Who is my **audience**?
2. What **question** am I trying to answer?
3. What **story** do I want people to walk away with?

1. Who is my audience?

Those who know the **ins and outs** of my research

Those who have **some knowledge** about my research

Those who **don't know anything** about my research!

1. Who is my audience?

Those who know the **ins and outs** of my research

Those who have **some knowledge** about my research

Those who **don't know anything** about my research!

Only me!



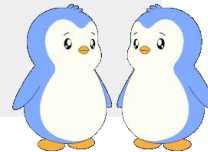
1. Who is my audience?

Those who know the **ins and outs** of my research

Only me!

Those who have **some knowledge** about my research

- PI / Committee members
- Collaborators
- Lab members
- Friends
- etc...



Those who **don't know anything** about my research!

1. Who is my audience?

Those who know the **ins and outs** of my research

Only me!

Those who have **some knowledge** about my research

- PI / Committee members
- Collaborators
- Lab members
- Friends
- etc...

Those who **don't know anything** about my research!

- Community members
- Study participants
- Conference attendees
- etc...



1. Who is my audience?

Those who know the **ins**
and outs of my research

Only me!

Those who have **some**
knowledge about my research

- PI / Committee members
- Collaborators
- Lab members
- Friends
- etc...

Those who **don't know**
anything about my research!

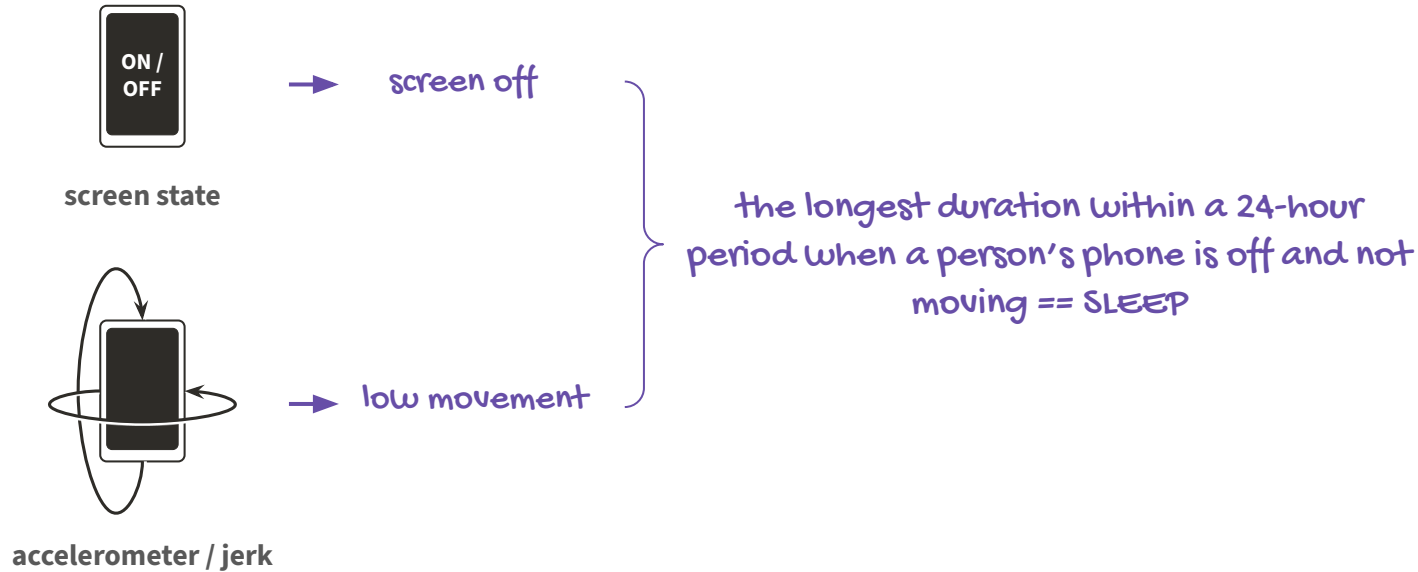
- Community members
- Study participants
- Conference attendees
- etc...



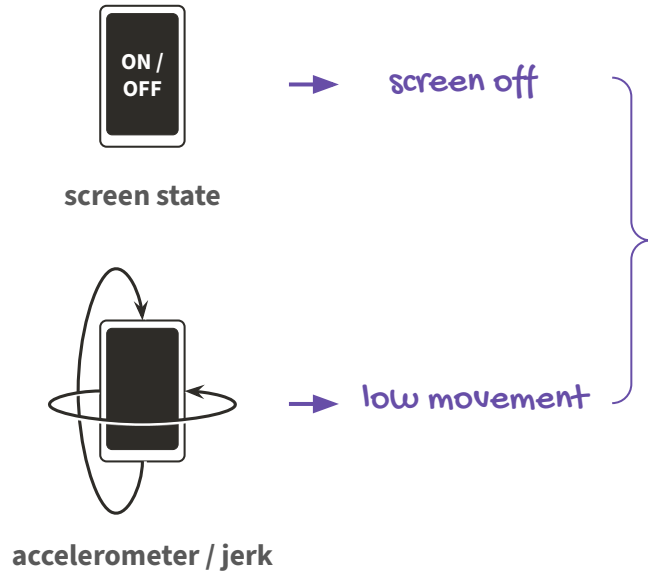
3 key questions

1. Who is my **audience**?
2. What **question** am I trying to answer?
3. What **story** do I want people to walk away with?

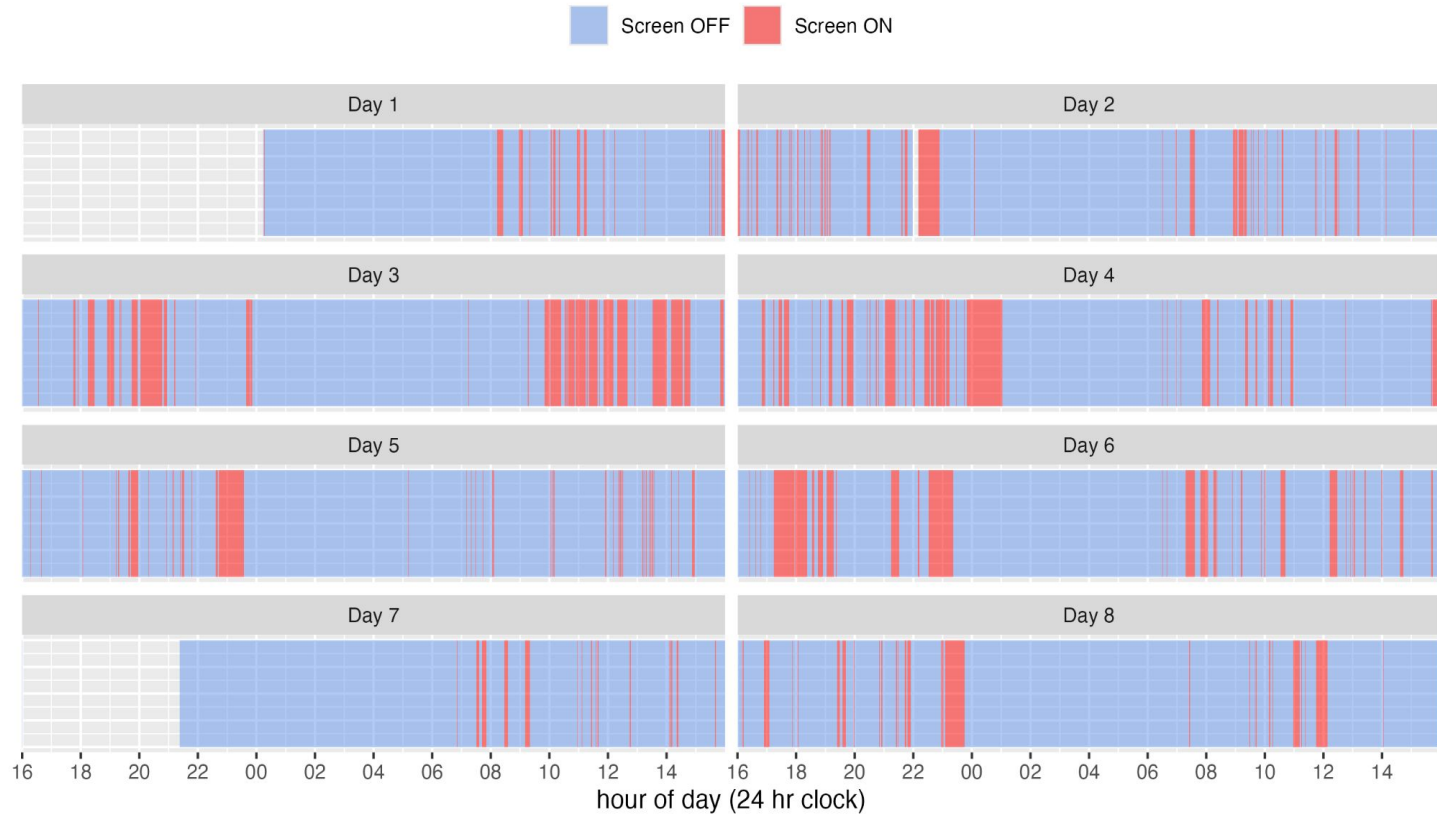
Using smartphone sensor data to capture sleep in young adults

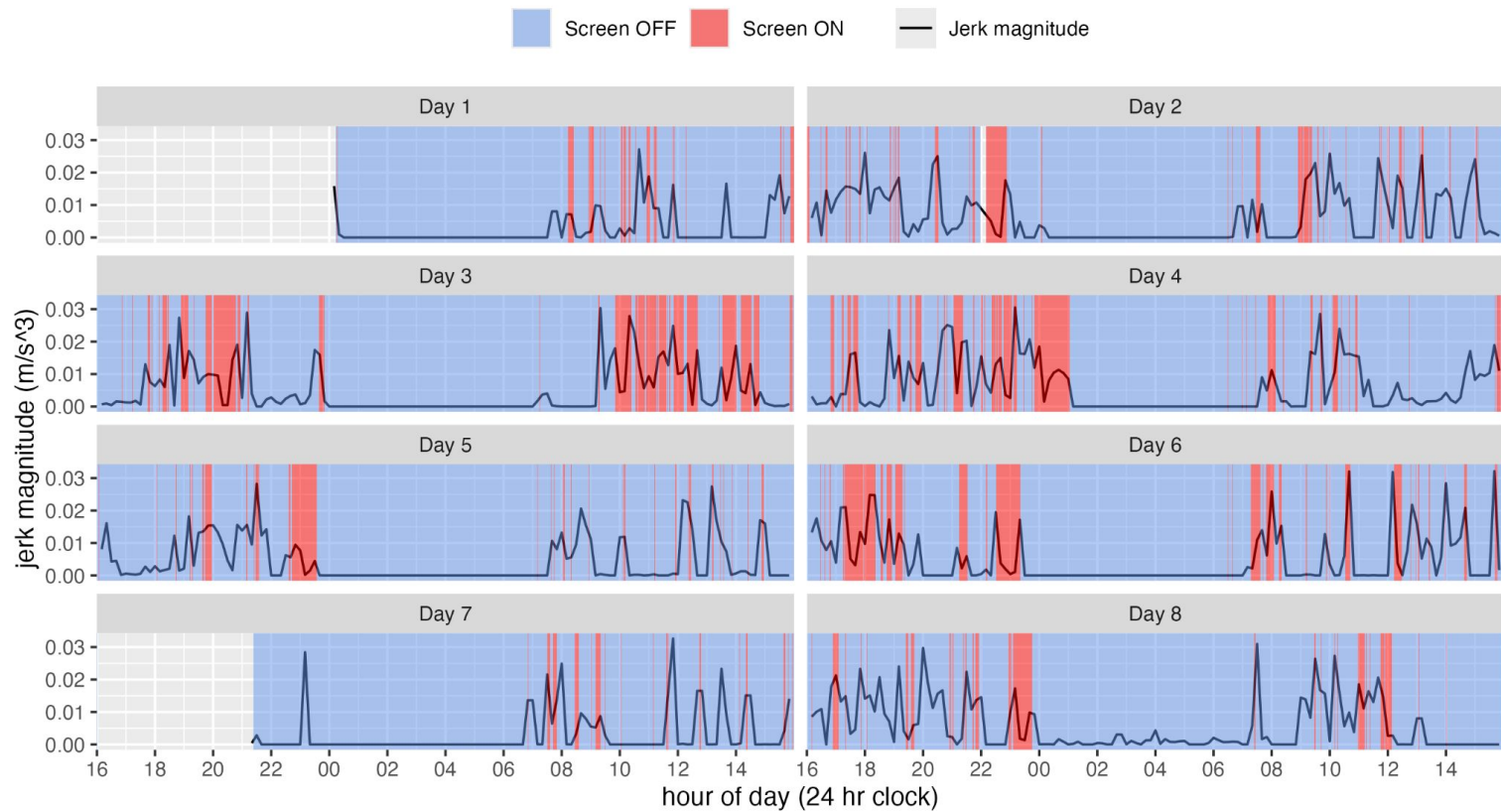


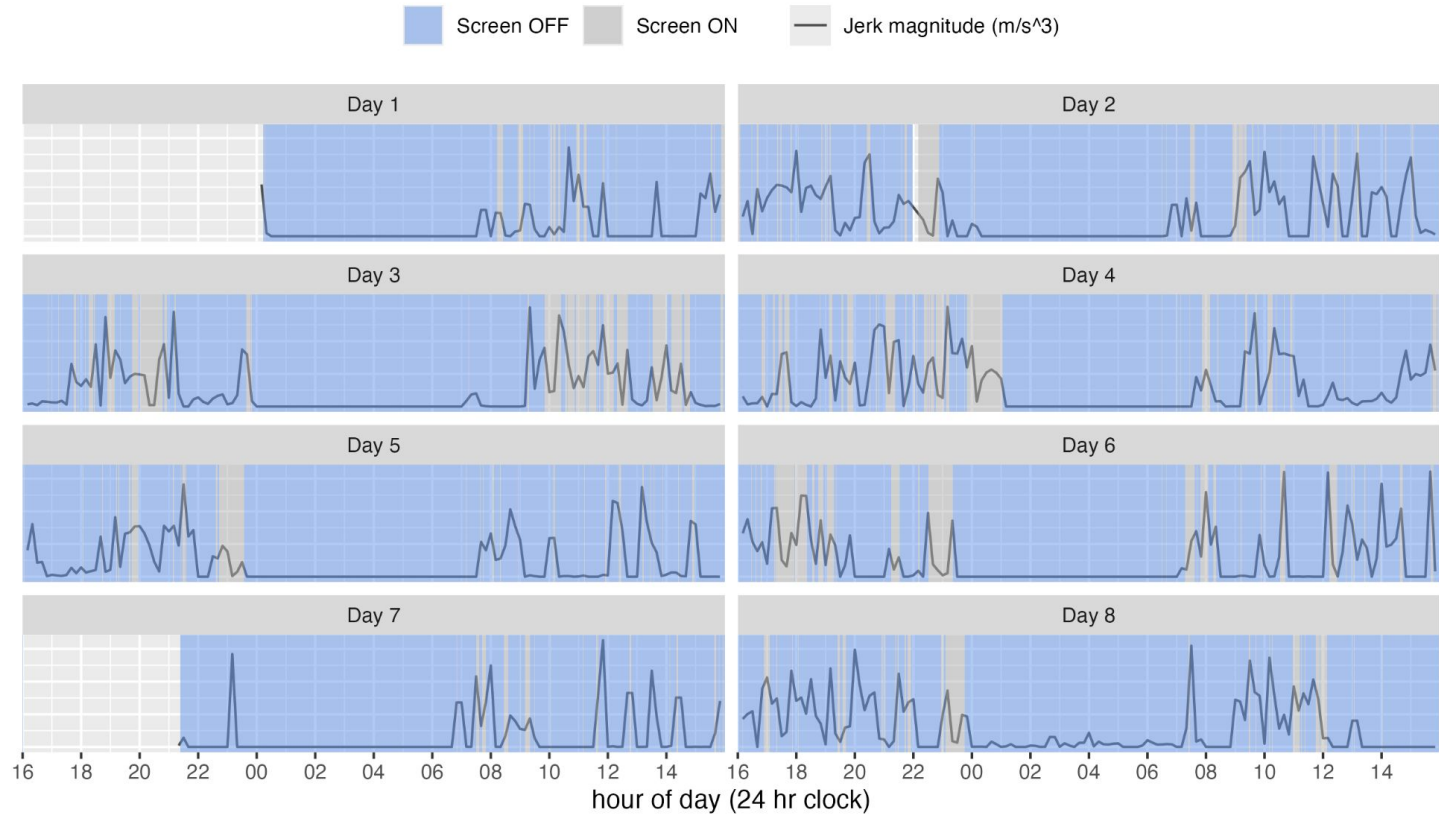
2. What question am I trying to answer?

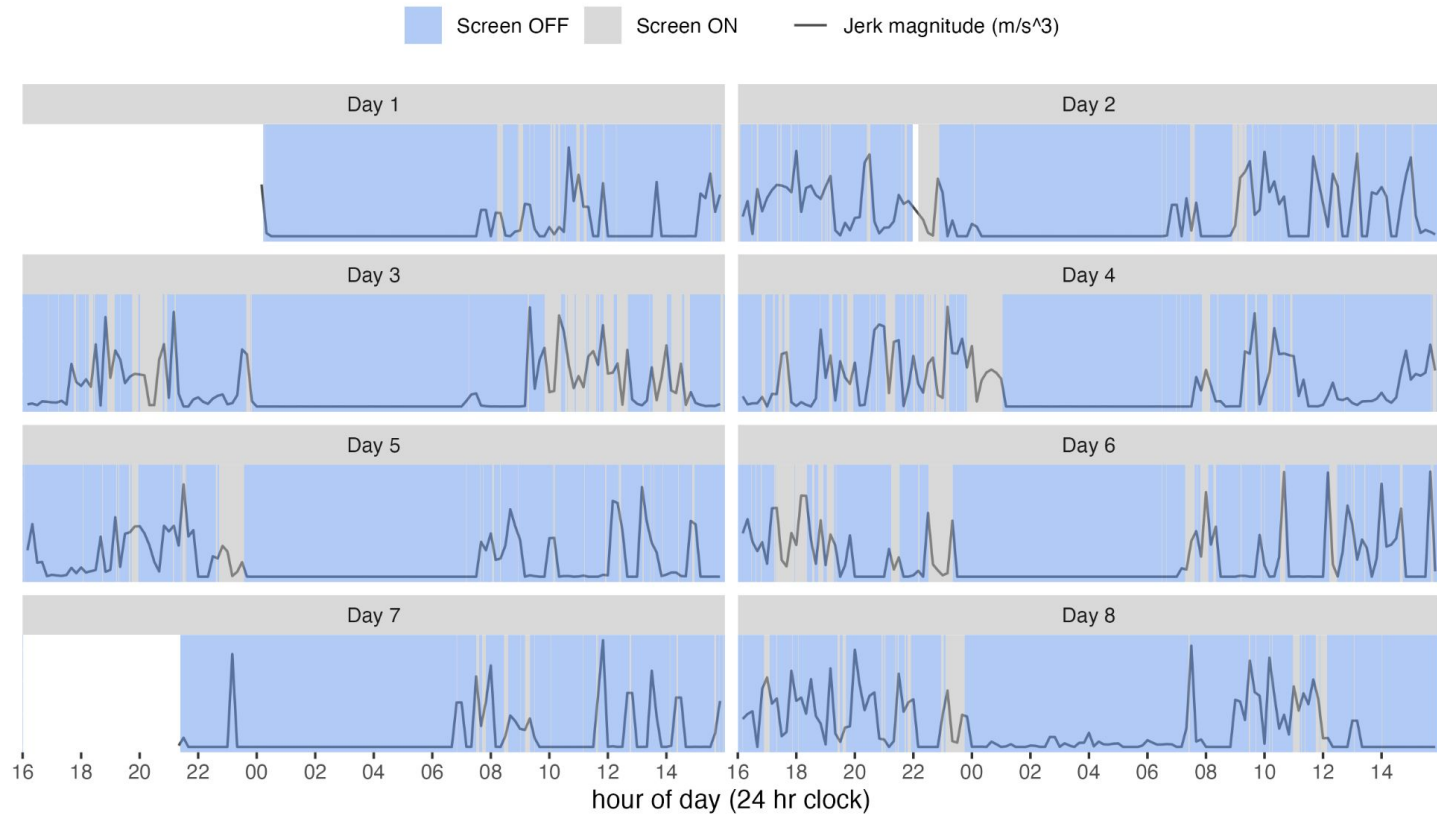


Question I want to answer: What are the **longest periods of time** when **screen = off** and **jerk < low movement threshold** per day?





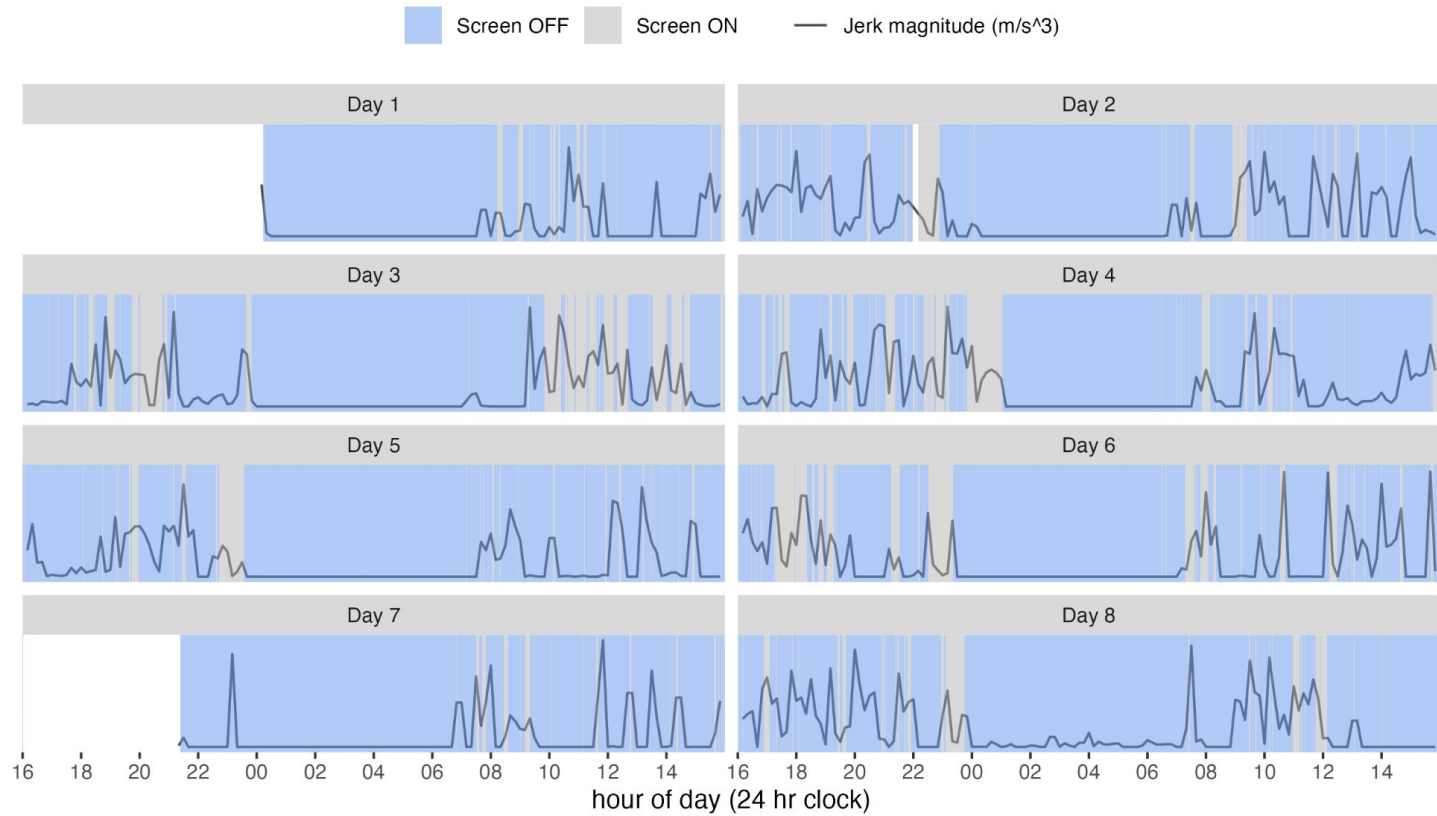


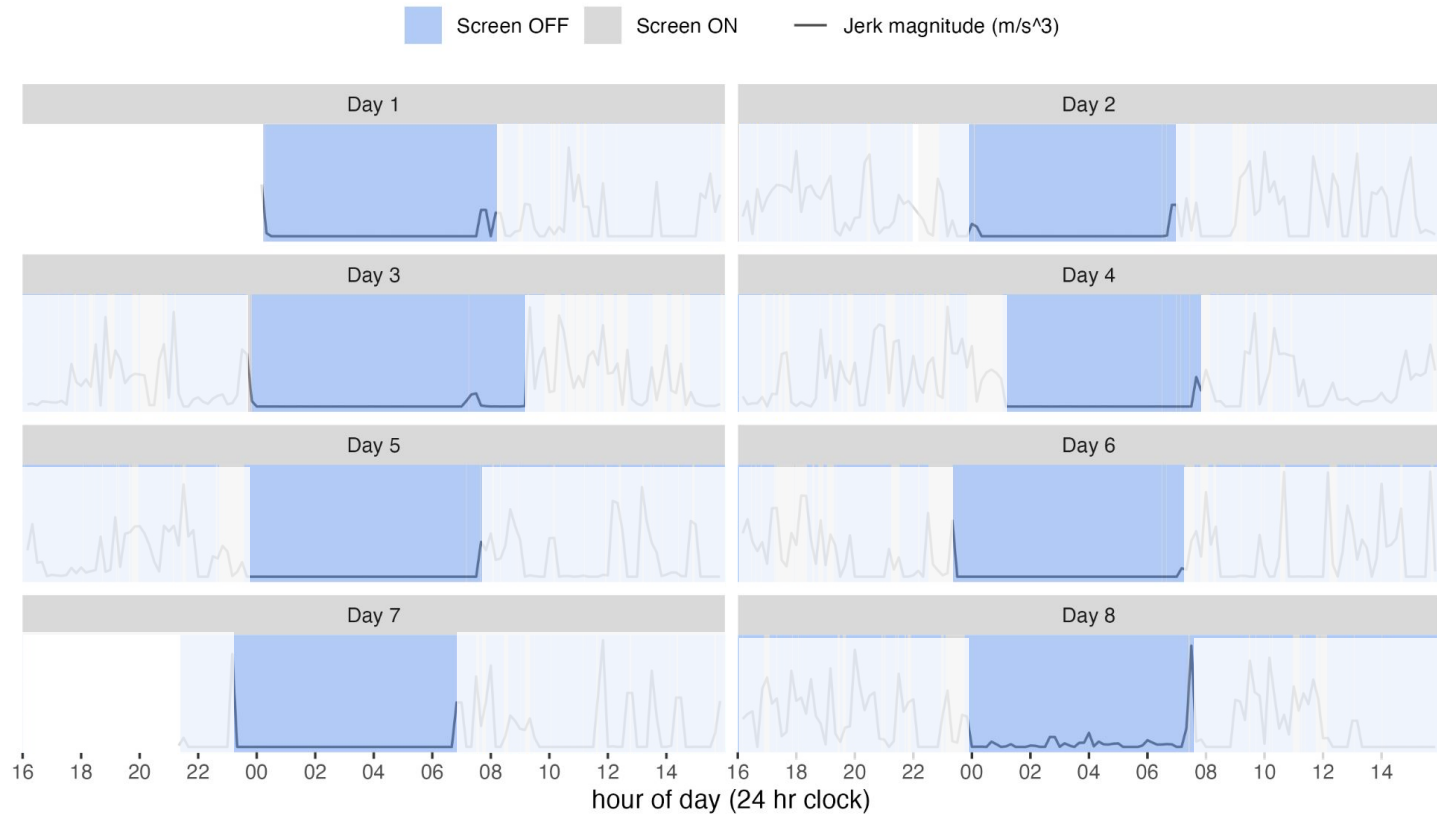


3 key questions

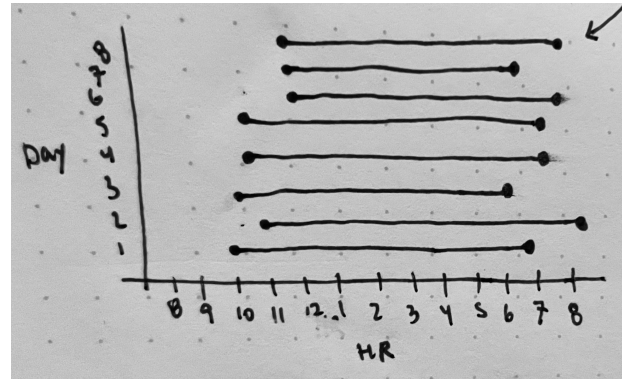
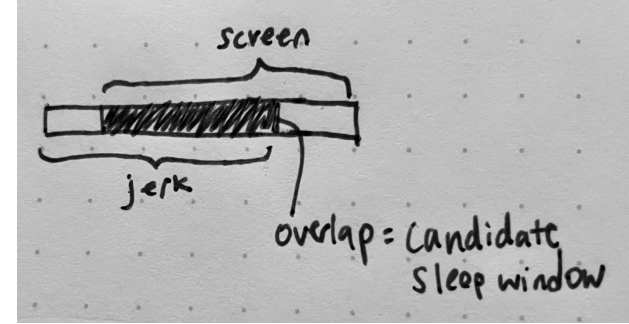
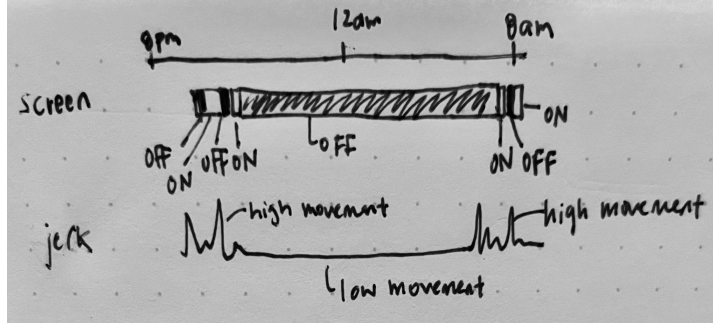
1. Who is my **audience**?
2. What **question** am I trying to answer?
3. What **story** do I want people to walk away with?

The overlap of sustained screen off and low phone movement within a 24-hour period may help infer when people go to bed and wake up

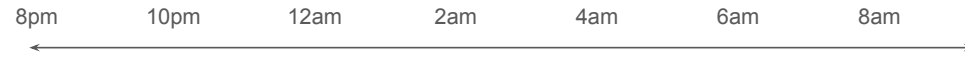




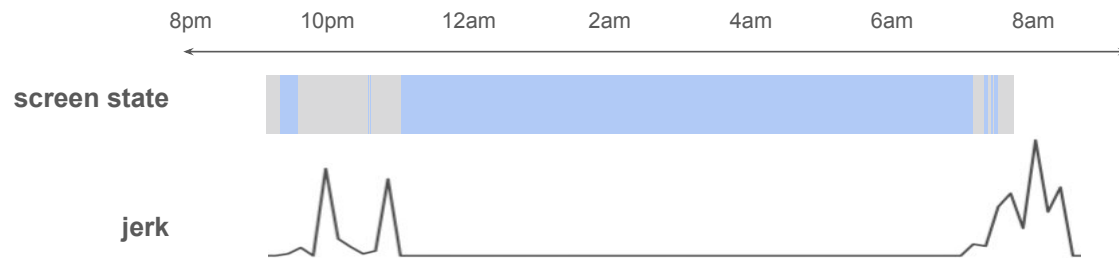
Some sketches



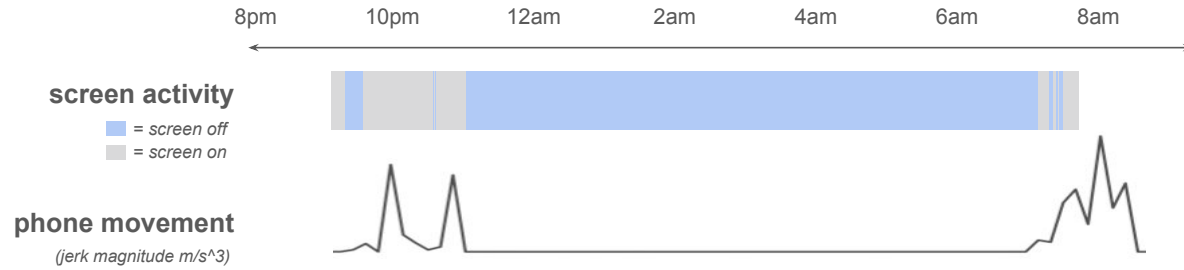
A.



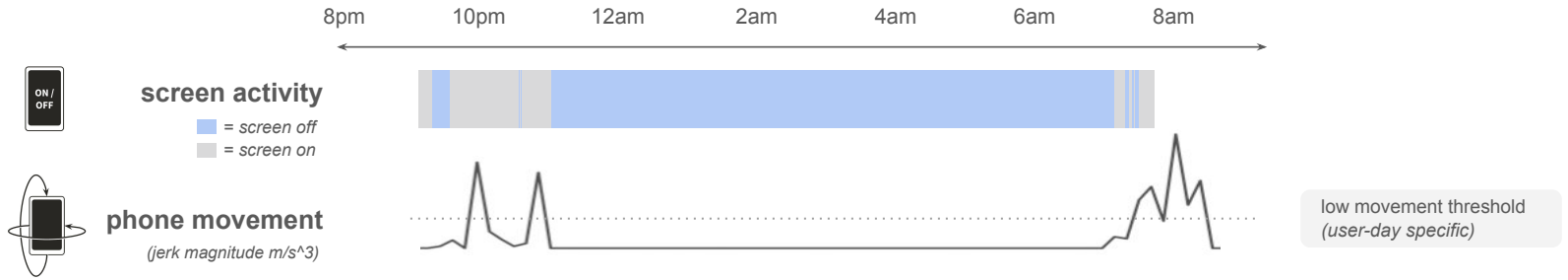
A.



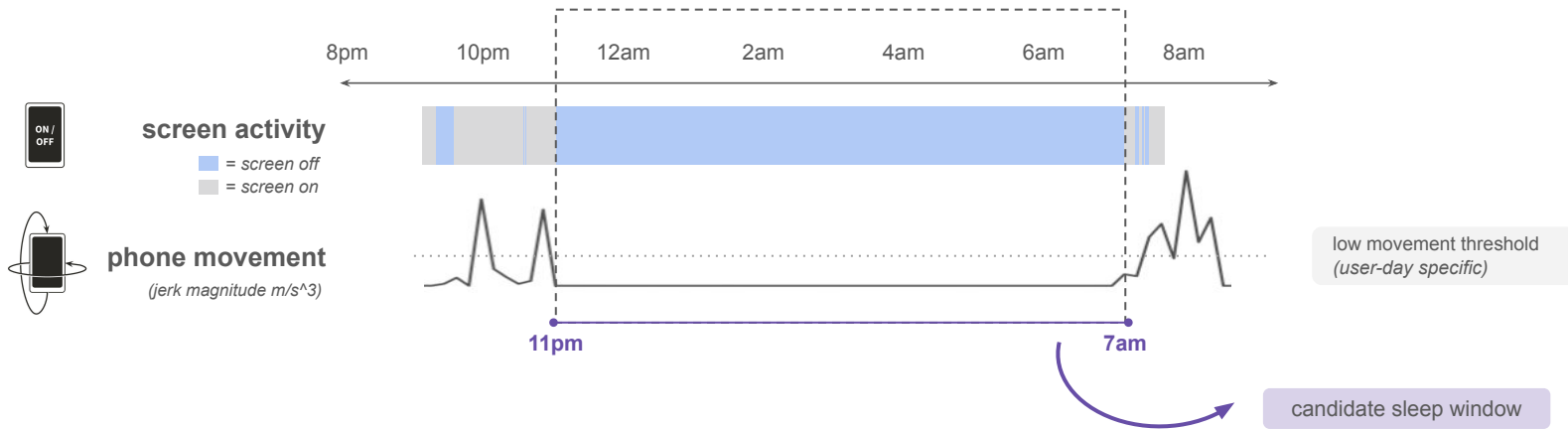
A.



A.



A.



A.



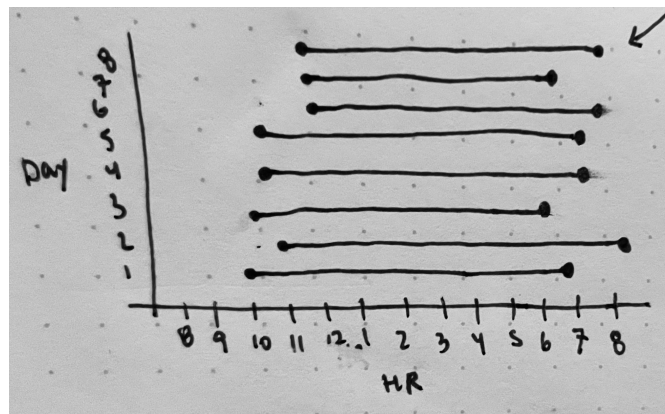
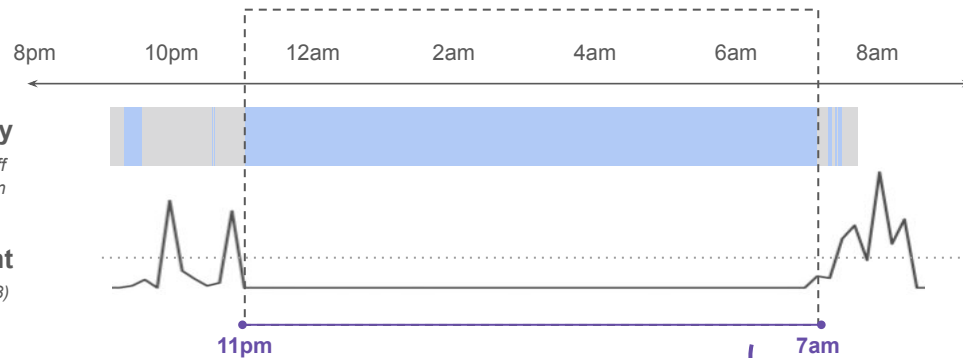
screen activity

■ = screen off
■ = screen on

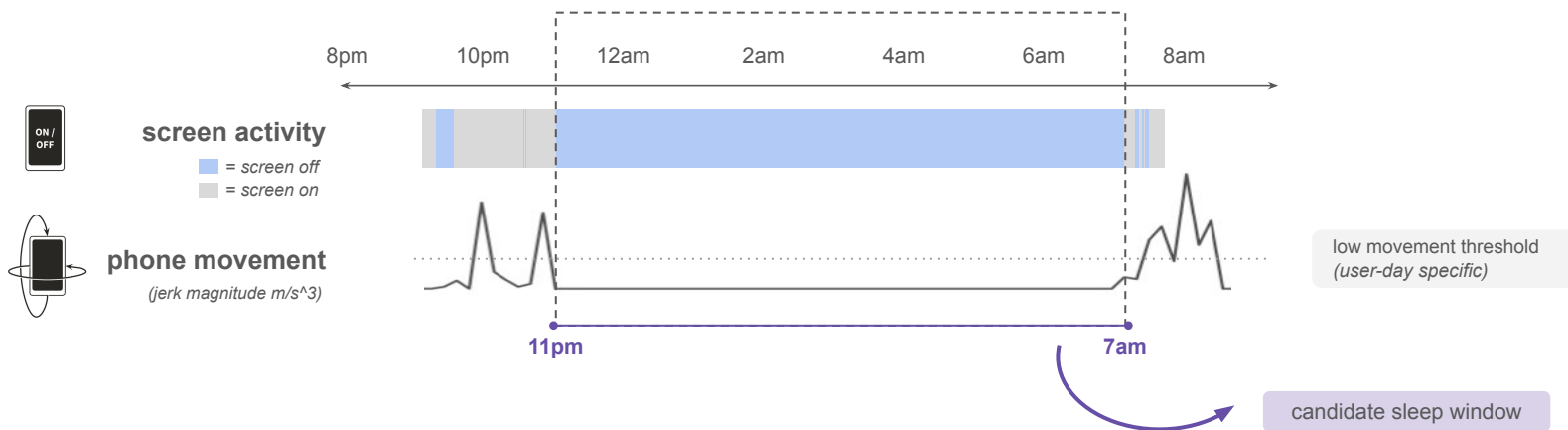


phone movement

(jerk magnitude m/s^3)



A.



B.

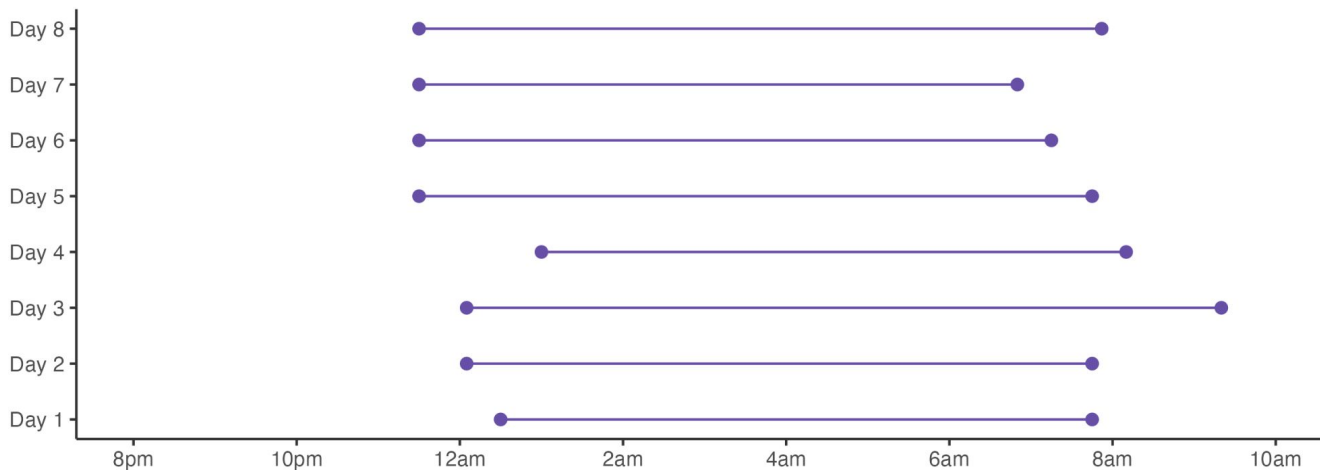


Figure. Estimated sleep and wake times from screen activity (screen state) and phone movement (jerk magnitude)

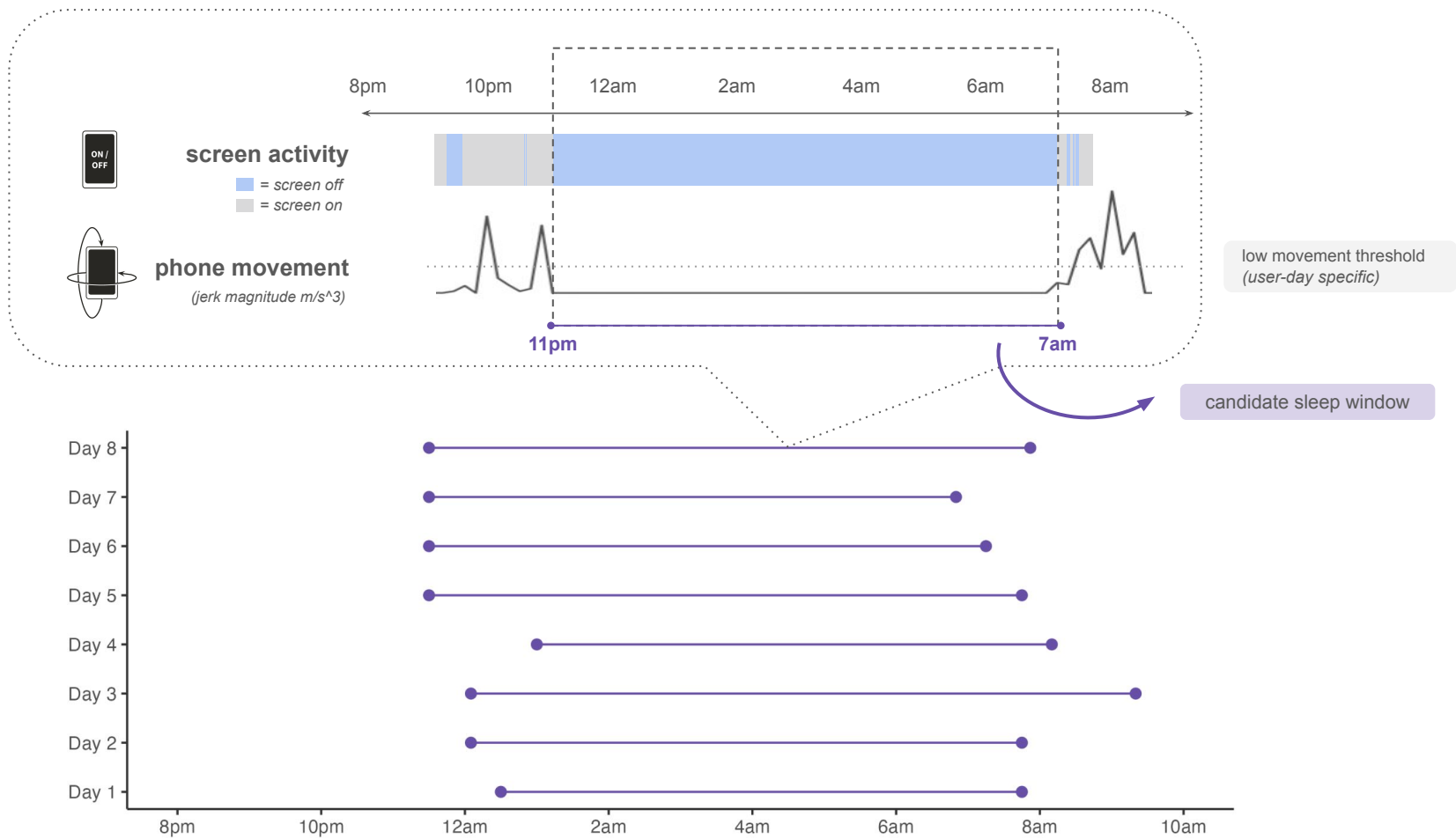
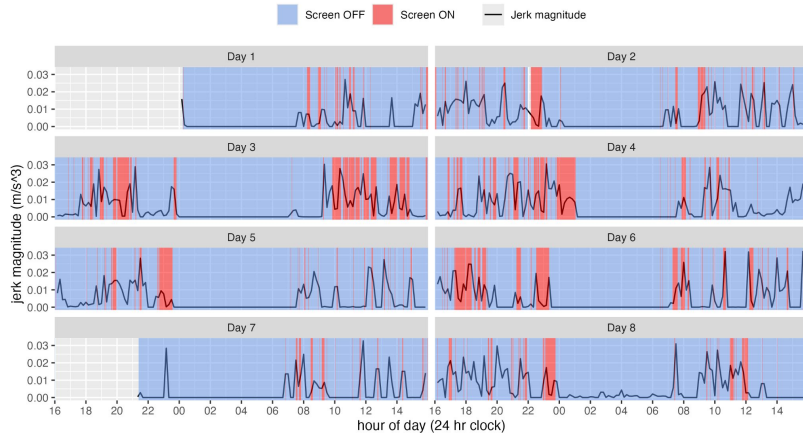
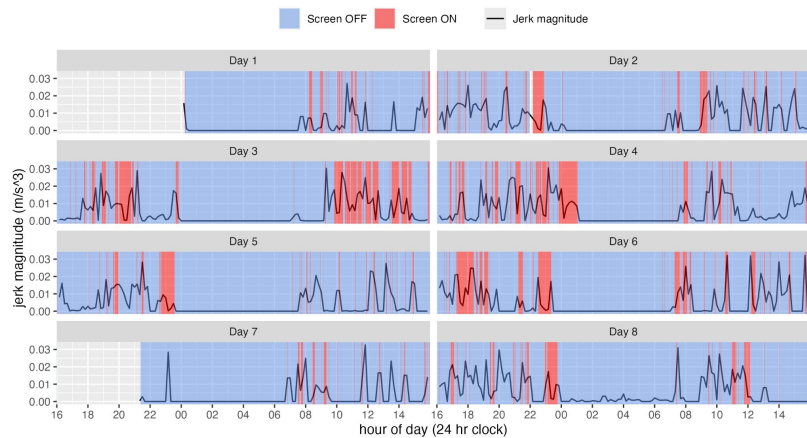


Figure. Estimated sleep and wake times from screen activity (screen state) and phone movement (jerk magnitude)

Before



Before



After

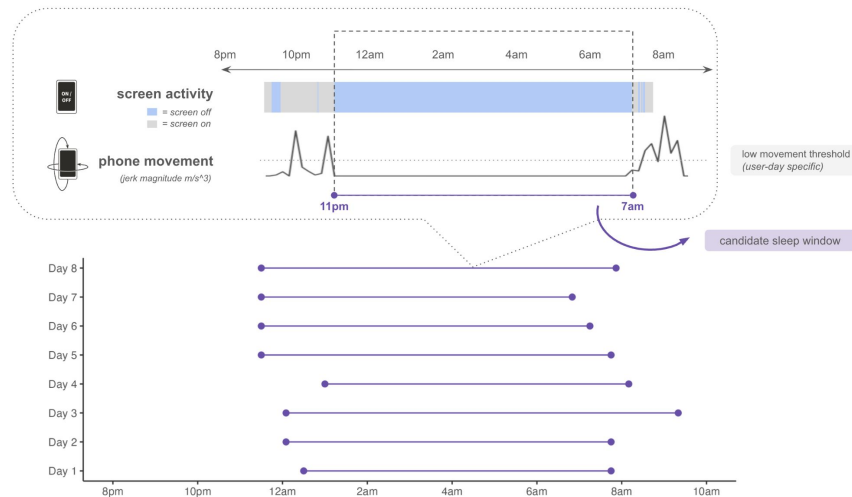


Figure. Estimated sleep and wake times from screen activity (screen state) and phone movement (jerk magnitude)

Where to go from here?

- Give it a try - it takes time, practice, and patience!
- If stuck, start sketching by hand; even if they're wonky
- Always keep the question and story in mind → this is your true north
- Cut unnecessary clutter
- Make details consistent
- Remember, small changes = big impact

Additional Resources

Nathan Yau — [FlowingData](#)

Cole Knafllic — [Storytelling with Data](#)

Edward Tufte — [The Visual Display of Quantitative Information](#)

ggplot2 — [R graph gallery](#)

Great color palette tools (some even color-blind friendly!) — [viridis\(\)](#),
[ColorBrewer\(\)](#)

Thank you!